

AIM:

Upload and analyse the given dataset and perform data preprocessing and visualisation.

PYTHON CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_excel("sales_data.xlsx")
print(df.head())
print(df.info())
print("\nMissing Values:")
print(df.isnull().sum())
print("\nDuplicate Rows:")
print(df.duplicated().sum())
plt.figure(figsize=(8, 5))
plt.scatter(
    df["Quantity"],
    df["Sales"],
    color="royalblue",
    edgecolors="black"
)
plt.title("Sales vs Quantity")
plt.xlabel("Quantity")
plt.ylabel("Sales")
plt.grid(True, alpha=0.3)
plt.show()

plt.hist(
    df["Sales"],
    bins=6,
    color="royalblue",
)
plt.title("Distribution of Sales")
plt.xlabel("Sales")
plt.ylabel("Frequency")
plt.grid(axis="y", alpha=0.3)
plt.show()
df.boxplot(
    column="Sales",
    by="Product",
    patch_artist=True,
)
plt.title("Sales Distribution by Product")
plt.suptitle("")
plt.xlabel("Product")
```

```
plt.ylabel("Sales")
plt.show()
```

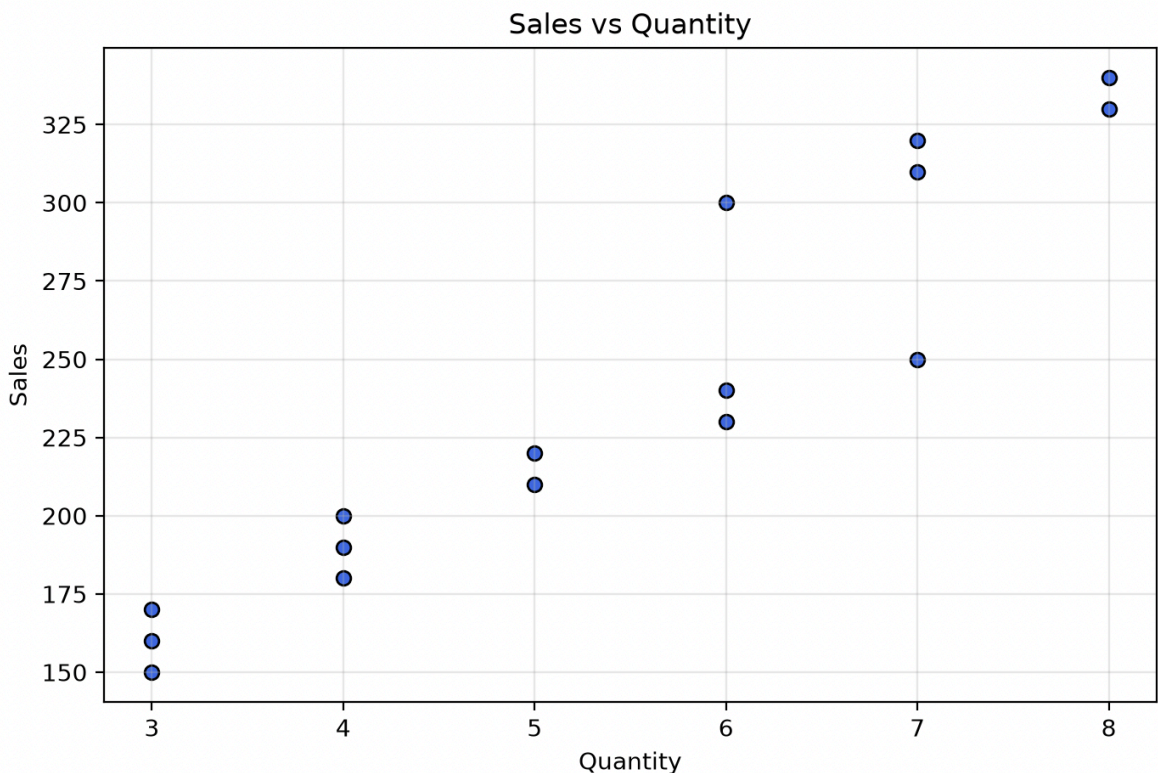
OUTPUT:

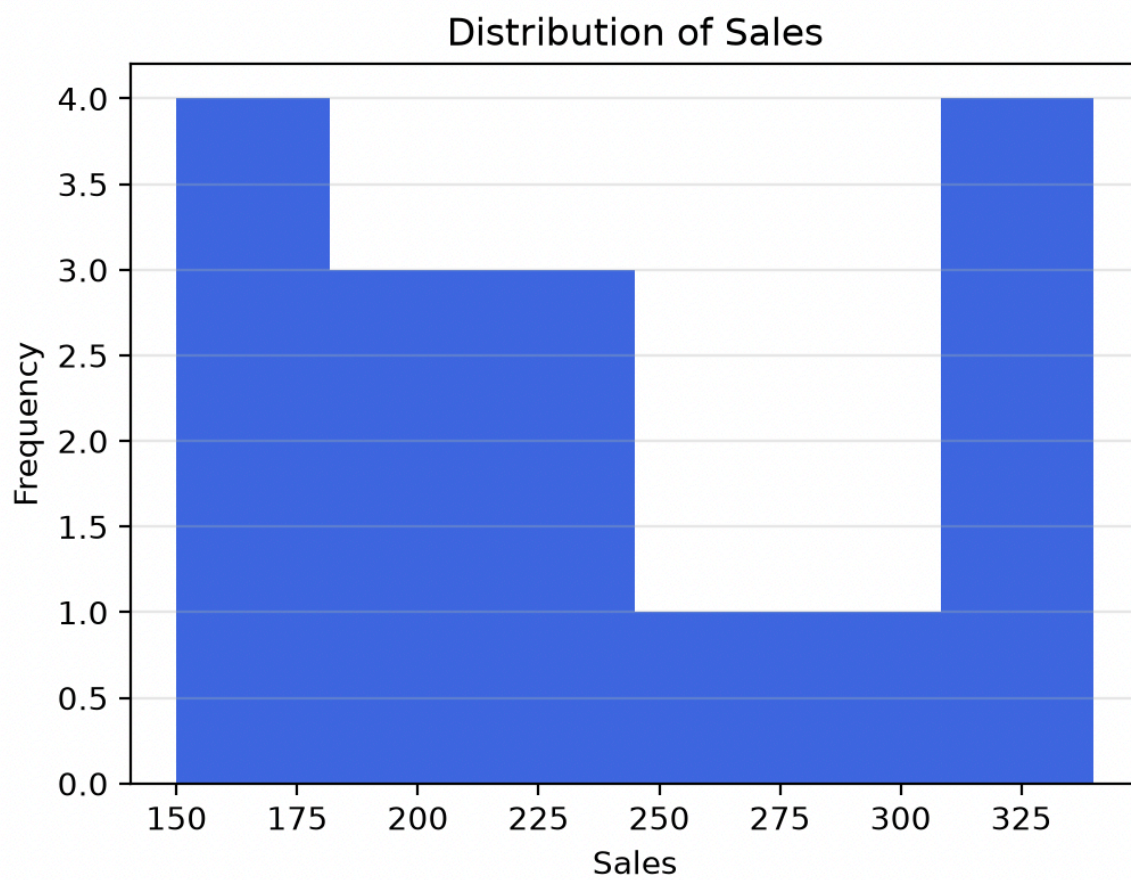
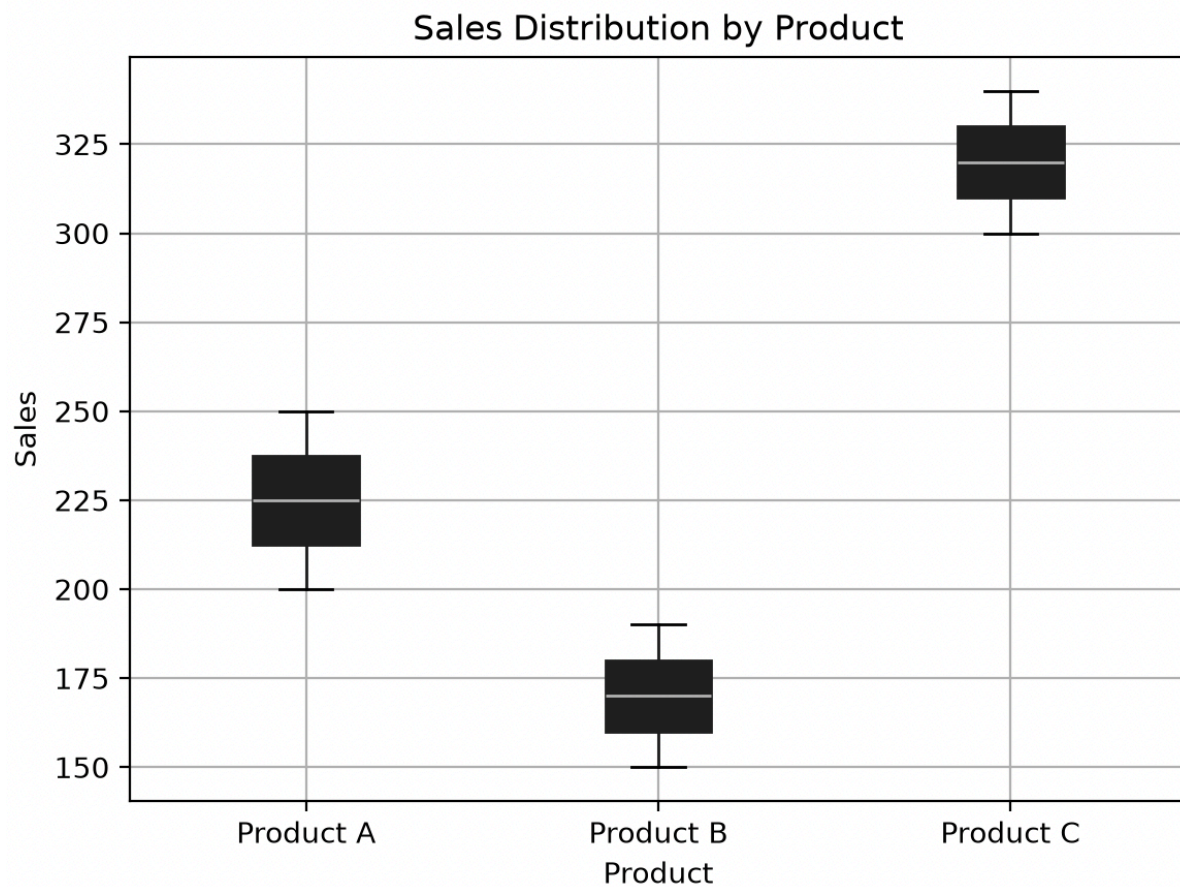
	Date	Product	Sales	Quantity	Region
0	2023-01-01 00:00:00	Product A	200	4	North
1	2023-02-01 00:00:00	Product B	150	3	South
2	2023-03-01 00:00:00	Product A	220	5	North
3	2023-04-01 00:00:00	Product C	300	6	East
4	2023-05-01 00:00:00	Product B	180	4	West

<class 'pandas.DataFrame'>
RangeIndex: 16 entries, 0 to 15
Data columns (total 5 columns):
Column Non-Null Count Dtype
--- ---
0 Date 16 non-null object
1 Product 16 non-null str
2 Sales 16 non-null int64
3 Quantity 16 non-null int64
4 Region 16 non-null str
dtypes: int64(2), object(1), str(2)
memory usage: 772.0+ bytes
None

Missing Values:
Date 0
Product 0
Sales 0
Quantity 0
Region 0
dtype: int64

Duplicate Rows:
0





RESULT :

The code is successfully executed.